

Advancing SME Growth through AI-Powered Feedback Analytics

A Master Thesis and Case Study of Tummy Clothing

A thesis submitted in partial fulfilment of the requirements for the

Master of Arts in Collective Intelligence

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Abiola Olanrewaju Ajao

Academic Supervisor: Professor Edmond Seabright

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Declaration on the Use of Artificial Intelligence

In accordance with UM6P and SCI guidelines on academic integrity and the transparent declaration of AI use, I acknowledge two forms of AI assistance used in preparing this thesis.

1. Platform development

The BizIntel Enterprise feedback analytics platform used Google's Gemini Pro large language model as its natural language processing engine. Gemini Pro was responsible for sentiment classification, thematic grouping, category assignment, and the generation of consumer experience summaries. The platform itself was conceived, designed, and deployed by the researcher, with Gemini Pro operating as the interpretation layer inside the analytics workflow. Its outputs were reviewed by the researcher before being included in this thesis.

2. Language refinement

A large language model was also used to refine and edit the written draft of this manuscript for clarity, coherence, and the linguistic standard expected of a master's thesis. All research design, data collection, analysis, interpretation, and conclusions remain the researcher's own intellectual work. All sources have been cited according to APA 7th edition conventions.

The dataset, platform outputs, findings, and recommendations presented in this study are the researcher's original contribution.

Signature: Abiola Olanrewaju Ajao

Date: July 8, 2026

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Abstract

Most small and medium-sized enterprises do not lack customer feedback. What they lack is the capacity to turn that feedback into structured intelligence for decision-making. The problem is sharper in fashion retail, where satisfaction depends on design, quality delivery and communication, fit, and the wider customer journey (Puccinelli et al., 2009; Verhoef et al., 2009). This thesis asks how collective intelligence can be applied through an LLM-supported feedback analytics platform to improve business decisions, using Tummy Clothing, a bespoke fashion SME, as a case study (Woolley et al., 2015; Riedl et al., 2021). Its main claim is that customer voice becomes a strategic asset once it is collected systematically, analysed with care, and it can be translated into insight that management can act on.

The research used BizIntel Enterprise, a custom feedback analytics platform powered by Google's Gemini Pro large language model, to capture structured survey responses and unstructured customer comments across several languages. The methodology combined quantitative measures, namely Customer Satisfaction (CSAT), Net Promoter Score (NPS), and Customer Effort Score (CES), with qualitative text analysis to produce a fuller picture of customer experience (Reis et al., 2022; Venkatesh, 2023). Because the customer base speaks English, Nigerian Pidgin, Yoruba, and Hausa, multilingual sentiment interpretation was required to read the comments accurately, and the study drew on established multilingual sentiment analysis research to support that work (Balahur & Turchi, 2014; Dashtipour et al., 2016; Koto et al., 2024). Drawing on 42 customer responses collected across WhatsApp, Instagram, and Facebook between May and June 2026, the findings show that collective intelligence can help SMEs shift from intuition-led management to evidence-based decision-making by surfacing recurring patterns in product quality, service responsiveness, sizing, and customer satisfaction.

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Chapter 1: Introduction

1.1 Background and Context

Contemporary business runs on data, digital communication, and rising customer expectations. For retail and fashion, firms are judged less by the products they sell in comparison to how well they understand and respond to customer experience (Verhoef et al., 2009; Puccinelli et al., 2009). Therefore, customer feedback analysis has become a strategic concern rather than just an administrative task. For small and medium-sized enterprises (SMEs), the difficulty is rarely a shortage of feedback. It is in turning scattered comments, complaints, and praise into clear insight that can guide both daily operations and long-term strategy (Puccinelli et al., 2009).

Fashion retail is an especially hard setting for this. Fashion products are often subjective, seasonal, and emotionally charged, so customers judge them not only on physical quality but on fit, style, packaging, delivery, communication, and how easy the brand is to deal with (Verhoef et al., 2009; Puccinelli et al., 2009). A well-designed garment can still leave a customer unhappy if it arrives late, fits poorly, or proves difficult to exchange. Fashion performance is therefore as much about experience quality as production quality (Puccinelli et al., 2009). For SMEs in fast-moving markets, customer feedback has to be treated as a core management resource rather than an afterthought.

The problem is more severe for SMEs in emerging markets. Most rely on informal channels: WhatsApp chats, Instagram direct messages, phone calls, and social media comments. Those channels hold valuable insight, but they are usually less organised in a way that lets managers review them systematically. The result is that owners operate with blind spots. They can see feedback without being able to detect which pain points recur most often, which issues matter most, or which operational problems sit just below the surface.

This thesis examines how collective intelligence can help close that gap. Collective intelligence here, means the capacity of structured systems to gather, organise, and interpret the voices of many customers so that a business can make better decisions. The idea has deep roots in decision theory and organisational research, which show that groups with access to diverse information can reach more accurate judgements than any single individual. The Condorcet Jury Theorem proved mathematically that the probability of a correct majority decision rises with the number of independent voters when each voter is more likely than not to be correct, approaching certainty as the group grows (Condorcet, 1785/1994). Later work on prediction markets extended the logic, showing that pooled information from many participants consistently outperforms expert judgement across a wide range of settings (Wolfers & Zitzewitz, 2004; Surowiecki, 2004). More recent studies of teams confirm that collective intelligence is a real, measurable property at the group level that predicts performance on many kinds of task (Woolley et al., 2015; Riedl et al., 2021). The BizIntel Enterprise platform puts this logic to work by aggregating and interpreting many customer voices, so that management can act on what the crowd is collectively saying rather than on the loudest or most recent complaint.

1.2 Problem Statement

Many SMEs collect feedback but never fully use it. Complaints land on different platforms, customer service or admin staff keeps informal records, and management hears only the loudest or most recent issues. The result is a visibility problem: the business receives feedback, but rarely in a form that makes real analysis possible. When that happens, recurring complaints such as sizing errors, slow response times, or poor product durability can go unnoticed even as many customers raise them.

The problem deepens when feedback is multilingual and culturally specific. Tummy Clothing's customers write out feedback in English, Nigerian Pidgin, Yoruba, and Hausa. A keyword-based review process cannot capture what such feedback actually means. Praise can carry local cultural weight, and frustration can be softened, coded, or expressed indirectly. A business that cannot read this language correctly risks misreading its customers entirely. Research on multilingual sentiment analysis confirms that low-resource and mixed-language contexts demand more careful modelling than standard monolingual text (Balahur & Turchi, 2014; Dashtipour et al., 2016; Koto et al., 2024).

There is a managerial dimension too. In a growing SME, leadership is often just the CEO and a few team leads, with little time or headcount to review every message by hand. This makes operational decisions rest on intuition, memory, or a handful of visible incidents rather than a complete view of customer reality. A gap opens between what the business believes is happening and what customers actually experience. This thesis addresses that gap by showing how an LLM-supported collective intelligence system can turn raw feedback into actionable knowledge.

1.3 Research Aim, Objectives, and Questions

This research examines how structured customer feedback and predictive analytics can improve decision-making and operational performance in a fast-paced fashion enterprise. It focuses on how collective intelligence can strengthen business processes when customer voices are captured systematically, analysed with care, and translated into strategic recommendations.

The study has four objectives:

- To deploy the BizIntel feedback analytics platform in a real business environment.
- To analyse multilingual customer feedback using natural language processing.

- To evaluate customer experience through CSAT, NPS, and CES measures.
- To turn feedback patterns into business intelligence that can guide executive decisions.

It is guided by three research questions:

- How do unstructured customer reviews relate to quantitative performance metrics such as CSAT, NPS, and CES?
- Can an LLM-supported collective intelligence platform interpret multilingual feedback and identify hidden operational issues?
- How does a feedback analytics dashboard help SME leadership move from reactive to evidence-based decision-making?

1.4 Significance of the Study

Academically, this research connects collective intelligence to business performance improvement, customer experience management, and applied natural language processing within a single case study. It adds to the growing discussion about how digital tools can support decision-making in real organisations, particularly in under-resourced settings where conventional analytics infrastructure is often unavailable (Woolley et al., 2015; Riedl et al., 2021).

Practically, the study offers an effective model for SME leaders who want to hear their customers more clearly without building a large technical department. It shows how feedback can be collected through simple digital surveys, processed with an LLM, and turned into an operational management dashboard. For the fashion business studied here, feedback becomes a continuous learning mechanism rather than a one-off survey exercise. The wider customer experience literature supports the view that businesses gain when they move customers from

satisfaction to loyalty and advocacy through better experience design (Verhoef et al., 2009; Puccinelli et al., 2009; Kamaladevi, 2009).

Chapter 2: Literature Review

2.1 Introduction

Customer experience has become central to modern retail because customers now judge businesses on the whole journey and not just the product alone. In retail fashion, that journey include style, fit, delivery, communication, packaging, and how easily problems get resolved. The literature treats customer experience management as a strategic factor that shapes loyalty, advocacy, and repeat purchasing (Verhoef et al., 2009; Puccinelli et al., 2009). For SMEs, the stakes are higher still, since limited staff and informal channels make it hard to track customer issues consistently. This chapter draws on work in retail strategy, collective intelligence, customer experience management, multilingual sentiment analysis, and fashion consumer behaviour to explain why feedback analytics matters for Tummy Clothing.

2.2 Customer Experience in Retail

Research on customer experience shows that the retail journey is shaped by many aspects, including promotion, product availability, service quality, logistics, and post-purchase interaction. Retail scholars have long held that experience is built over time rather than in a single moment, which means businesses must attend to the full sequence of interactions rather than the final sale alone (Verhoef et al., 2009). In fashion, a customer can be pleased with one part of the journey and frustrated by another; a beautiful design can still disappoint if delivery drags or sizing is wrong.

Studies of retail customer experience also stress that firms should think in journeys, not isolated transactions. Customers do not split a brand into internal departments. They experience it as one continuous relationship, which makes consistency essential. A company strong in product

design but weak in response time still delivers a poor experience overall. For that reason, customer experience management has become a central concern in retail strategy (Puccinelli et al., 2009; Kamaladevi, 2009). Firms that understand this are more likely to convert satisfaction into loyalty, and loyalty into advocacy.

This literature bears directly on the present study because Tummy Clothing operates in a setting where product quality and service quality is of high importance. The feedback collected through BizIntel shows that customers do not talk about clothing in isolation; they tie garment quality to delivery, communication, packaging, and trust. That is why customer experience offers the right theoretical lens for this case.

2.3 Collective Intelligence and Organisational Learning

Collective intelligence refers to the capacity of groups to produce knowledge stronger than any individual could generate alone. In organisations, the idea is usually tied to collaboration, shared information, and better decision-making. Research on teams and organisations suggests that collective intelligence is not simply a matter of involving many people; it depends on how information is structured, shared, and interpreted (Woolley et al., 2015; Riedl et al., 2021). When that process works, organisations respond more intelligently to uncertainty.

The theoretical foundation traces back to the Condorcet Jury Theorem, which showed that when each member of a group is independently more likely than not to be correct, the probability that the group majority reaches the right conclusion always exceeds that of any single individual and approaches certainty as the group grows (Condorcet, 1785/1994). The same principle was later applied to prediction markets, where aggregated bets from many participants have outperformed expert panels across domains from elections to commodity prices (Wolfers & Zitzewitz, 2004; Surowiecki, 2004). In organisational settings, Woolley et al. (2015) and Riedl

et al. (2021) found that teams exhibit a measurable collective intelligence factor, distinct from the average or maximum ability of their members, that predicts performance across many tasks. Applied to business, this means the aggregated judgement of many customers, once properly structured, can reveal truths about a firm's performance that no single manager or reviewer would spot alone.

This study uses collective intelligence in a practical sense. Customer feedback is treated as a distributed source of knowledge, and the BizIntel platform is the mechanism that structures that knowledge into a form management can act on. The aim is not to replace judgment with automation but to support organisational learning by showing leaders what customers are collectively saying (CoachHub, 2026).

2.4 Feedback Analytics and Customer Insight

At its simplest, feedback analytics collects customer responses and converts them into insight, usually by pairing quantitative scoring with qualitative interpretation. Literature increasingly favours this integrated approach because numbers alone do not explain behaviour and comments alone do not reveal scale or pattern. A mixed method enables the feedback system to give a more balanced view of performance (Chattermill, 2026; Verhoef et al., 2009). Analytics tools help manage the overload of digital-commerce feedback by grouping complaints, identifying themes, and flagging recurring issues. Often their value lies less in reporting what customers like or dislike than in indicating what the business should do next (Chattermill, 2026).

2.5 Fashion, Identity, and Consumer Behaviour

Fashion is not only about clothing. It is also about identity, self-expression, belonging, and social meaning. Consumers use fashion to signal who they are or how they wish to be seen,

which makes fashion purchases more emotional and socially driven than most others (Kotler & Keller, 2016). Because fashion is symbolic, customers are often more sensitive to detail: fit, stitching, texture, and presentation can all decide whether a product feels premium or disappointing. The literature shows that success in fashion retail rests not only on product appeal but on consistency in service and delivery (Puccinelli et al., 2009; Klaus & Kuppelwieser, 2023). Feedback matters here because customer reactions expose hidden gaps between expectation and experience.

2.6 Multilingual Sentiment Analysis

Language diversity is a major challenge in customer feedback analysis. Standard sentiment systems tend to perform well only on clean, monolingual text. In multilingual environments, and especially in under-resourced language contexts, classification becomes harder because a single message can carry several languages and culturally specific meanings (Balahur & Turchi, 2014; Koto et al., 2024). Research shows that low-resource languages need particular care because they lack the large datasets and language tools available for English (Mabokela et al., 2023; Dashtipour et al., 2016). This study met that limitation directly: one entry written in mixed Yoruba and English was misclassified by the LLM before the researcher's contextual review caught and corrected the error.

2.7 Business Intelligence, Strategy, and Decision-Making

Business intelligence earns its keep when it supports decisions rather than simply reporting data. In retail, that means using customer insight to sharpen product planning, service quality, inventory decisions, and communication strategy. A business that understands its customers can respond faster, reduce friction, and build loyalty (CoachHub, 2026; Chattermill, 2026). This matters especially for SMEs, which usually cannot match the analytic depth of large

corporations. They need simple but effective tools that help them choose well with limited resources, and the platform does more than report whether customers are satisfied; it helps management see the reasons behind those judgments.

2.8 Research Gap

Despite a strong literature on customer experience, collective intelligence, and sentiment analysis, a gap remains around applying these ideas in small fashion businesses in emerging markets. Most existing work centres on large firms, general retail, or technical models rather than the practical realities of SMEs with multilingual customer bases. Retail customer experience research has largely developed through studies of established Western retailers and department stores (Verhoef et al., 2009; Puccinelli et al., 2009). Sentiment analysis research has advanced through English-language benchmark datasets such as product review corpora and social media samples (Dashtipour et al., 2016; Pradha et al., 2019). Commercial feedback platforms like those documented by Chattermill (2026) and Qualtrics (2025) are built for enterprise clients with dedicated analytics teams. None speaks directly to a small bespoke fashion brand serving a code-switching customer base across two continents.

A second gap concerns the integration of structured and unstructured feedback. Few studies bring the two together in a real organisational setting, yet that combination is exactly what managers need to understand not only how customers feel but why. A third gap concerns multilingual, culturally embedded feedback, which calls for more inclusive analytical approaches than English-only systems can offer (Balahur & Turchi, 2014; Koto et al., 2024; Mabokela et al., 2023). This study addresses all three: it applies established customer-experience and collective-intelligence ideas to a small bespoke fashion brand rather than a large enterprise, it combines structured scores with unstructured comments in a single working

system, and it treats multilingual, code-switched feedback as analysable data rather than noise to be discarded.

2.9 Chapter Summary

Across these five bodies of work, one argument holds the chapter together: customer feedback is a strategic resource capable of improving products, services, and decisions when it is gathered and interpreted well. The collective intelligence tradition, running from Condorcet's jury theorem through prediction markets to modern team research, provides the theoretical basis for expecting that aggregating customer voices yields more reliable insight than any single perspective. The next chapter introduces the background of the case study, and Chapter 4 sets out the methodology and system design.

Chapter 3: Background of the Case Study

3.1 Introduction

This chapter sets out the organisational context in which the BizIntel platform was deployed and traces how the case study moved from data collection to real business application. A case study approach suits the research because the concern is context, process, and practical change rather than abstract measurement (Yin, 2014). In retail research, customer experience is often best understood as part of the full buying journey rather than a single moment of satisfaction (Puccinelli et al., 2009; Verhoef et al., 2009).

The chapter also links the study to collective intelligence. The main idea is that useful organisational knowledge can emerge when many customer voices are gathered, structured, and interpreted together. In this sense the platform acts as a bridge between distributed customer experience and managerial decision-making. Research on collective intelligence in organisations supports the view that groups can outperform individuals when information is well organised and used for coordination and problem-solving (Woolley et al., 2015; Hevner et al., 2004).

3.2 Tummy Clothing as a Case

Tummy Clothing is a bespoke fashion business that grew from a local tailoring operation into a brand serving customers across Nigeria and abroad. Its reach extends to the United Kingdom, making it both a local and a cross-border brand. That kind of growth brings opportunity but also operational pressure. A business once built on close personal contact with a small group of clients must now manage a wider base, with more service expectations and more complex logistics.

Like many SMEs, the company runs a thin structure. Production, quality control, customer communication, and logistics sit with a small team. The customer experience literature shows that brand performance depends on many connected elements, including product quality, delivery, communication, and after-sales support (Puccinelli et al., 2009; Kamaladevi, 2009). For Tummy Clothing, then, feedback could not be a side issue. The brand made a strong case for feedback analytics: it already had customer attention and market demand but lacked a single structured view of what customers were saying.

3.3 Operational Setting Before the Platform

Before the platform, customer feedback was spread across WhatsApp, Instagram direct messages, email, and informal conversation. The messages were useful, but they were not stored in a way that let the customer service admin spot recurring patterns quickly. A quality control officer might notice a single complaint about sizing or a single compliment about design while the wider pattern stayed hidden.

This kind of data fragmentation is common in SMEs. The information exists but is not centralised, so decisions rest on memory, personal impression, or isolated incidents rather than a full reading of customer voice. That is risky, because retail performance depends on consistency. A business can look successful while repeated problems such as delivery delays, incorrect sizing, or product defects continue unnoticed (Verhoef et al., 2009; Klaus & Kuppelwieser, 2023). The setting was further complicated by the fact that the company served both local and international customers with different service expectations, which made the BizIntel deployment especially important for creating segmented visibility.

3.4 The BizIntel Enterprise Platform

BizIntel Enterprise was introduced as a structured feedback analytics tool for Tummy Clothing. Its job was to collect survey responses, organise them, classify them, and present the results in a form management could use. The system gathered numerical ratings and free-text comments alike, making it possible to compare the overall level of satisfaction against the reasons behind it.

The platform is built around four connected modules. The first is a Consumer Insights Dashboard that consolidates Gemini Pro-processed feedback into an immediate view of product and service health, showing complaint volume, compliment volume, hidden opportunities, and narrative descriptions of the consumer experience. The second is a set of advanced data visualisations built with the Recharts library, including donut and pie charts for sentiment distribution, line charts to track issue frequency over time, stacked bar charts for category comparison, and geospatial mapping. The third is an automated SWOT Grid that maps Gemini Pro's thematic summaries onto a Strengths, Weaknesses, Opportunities, and Threats matrix, with a linked Contention Analysis. The fourth is an Export Action Bar that lets managers export charts and summaries into PDF and CSV reports for internal sharing.

Plate 2 shows the Feedback Records dashboard, which displays all 42 entries with their date, source, sentiment, category, and message fields. Plate 3 shows Master Data Hub, which stored customer Metadata securely and allowed the team to segment responses by location and capture date. Plate 4 shows the survey form as seen by customers.

Feedbacks

Manage and review all customer responses

Export All (CSV)

All Forms

Search feedback...

Date	Source	Sentiment	Category	Message	Action
Jun 30, 2026, 05:35 PM	Rate our Groom's Styling	POSITIVE	FASHION/APPEL	Omo, this cloth choke! E fit me well well. Everybody dey ask me where I sew am. Una do well for Tummy Apparel. Summary: The customer is extremely satisfied with their purchase from Tummy Apparel, praising the quality and fit of the cloth.	View Reply
Jun 30, 2026, 05:32 PM	Rate our Groom's Styling	MIXED	FASHION/APPEL	Kayan yana da kyau, but the neck of the senator wear is too tight. Make sure you check the measurements well before shipping next time. Summary: The customer likes the product quality but experienced sizing issues with the neck of the senator wear, suggesting a need for improved measurement checking.	View Reply
Jun 30, 2026, 05:30 PM	Rate our Groom's Styling	NEGATIVE	LOGISTICS AND CUSTOMER SUPPORT	O ti su mi o. I paid for express delivery to Ibadan since Monday, up till now nothing. Your customer care is just ignoring my messages. Summary: Customer expresses dissatisfaction with express delivery service to Ibadan and unresponsive customer care.	View Reply
Jun 30, 2026, 04:55 PM	Rate our Groom's Styling	POSITIVE	FASHION	Absolutely smashing! The two-piece set arrived in London just in time for my event. Quality is 10/10. Summary: The customer is extremely satisfied with their two-piece set purchase, praising its high quality and timely arrival.	View Reply
Jun 30, 2026, 04:52 PM	Rate our Groom's Styling	POSITIVE	FASHION AND APPAREL	Make I no lie, Tummy Apparel sabi this work. The agbada make me look like chief for London streets. Respect! Summary: The customer is extremely satisfied with Tummy Apparel's agbada, which makes them look stylish and respectable on London streets	View Reply
Jun 30, 2026, 04:47 PM	Rate our Groom's Styling	NEGATIVE	ORDER FULFILLMENT AND CUSTOMER SUPPORT	Ewo ni gbogbo radarada yi? I ordered the blue Ankara but you sent me green, yet nobody is answering my DMs. Summary: Customer received incorrect order and is experiencing difficulty getting assistance from customer support	View Reply
Jun 30, 2026, 04:45 PM	Rate our Groom's Styling	POSITIVE	CUSTOMER SERVICE	Top-notch customer service! The rep patiently helped me pick the right size for my traditional wedding. highly recommend Tummy Apparel. Summary: Tummy Apparel provided exceptional customer service for a traditional wedding, exceeding expectations with patient and knowledgeable support.	View Reply
Jun 30, 2026, 04:42 PM	Rate our Groom's Styling	POSITIVE	PRODUCT AND DELIVERY SATISFACTION	Gaskiya naji dadin kayan nan. It fits perfectly and the delivery to Kano was faster than I expected. Summary: Customer is extremely satisfied with the product fit and fast delivery to	View Reply

Plate 1: BizIntel feedback records dashboard: date, source, sentiment, category, message, and action fields across all 42 submissions

☐ Show At-Risk Only Search leads... Export	
Location	Captured On
Nigeria Enugu, N/A	Jun 30, 2026
Nigeria Abuja, N/A	Jun 30, 2026
Nigeria Lagos, N/A	Jun 30, 2026
United Kingdom London, N/A	Jun 30, 2026
Nigeria Rivers, N/A	Jun 30, 2026
Nigeria Lagos, N/A	Jun 30, 2026
Nigeria Enugu, N/A	Jun 30, 2026
Nigeria Kano, N/A	Jun 30, 2026
Nigeria Lagos, N/A	Jun 30, 2026
United Kingdom London, N/A	Jun 30, 2026
Nigeria Ogun, N/A	Jun 30, 2026
Nigeria Abuja, N/A	Jun 30, 2026

Plate 2: Master Data Hub showing location and capture date of feedback submissions, segmented by geography.

The screenshot displays a feedback form titled "Rate our Groom's Styling" within the BizIntel Platform. The form is designed to collect customer feedback on a groom's styling service. It includes three primary rating scales: CSAT (Customer Satisfaction), NPS (Net Promoter Score), and CES (Customer Effort Score). The CSAT scale ranges from 1 (Very unsatisfied) to 5 (Very satisfied), with a rating of 4 selected. The NPS scale ranges from 0 (Never recommend) to 10 (Absolutely recommend), with a rating of 7 selected. The CES scale ranges from 1 (Very Difficult) to 5 (Very Easy), with a rating of 4 selected. Below the rating scales is an open-text field labeled "Tell us more *" for additional feedback. A "Continue" button is located at the bottom of the form.

Plate 3: BizIntel Platform's feedback form overview showing CSAT, NPS, CES scales and open-text field.

3.5 Geographic and Language Context

The customer base was geographically wide. Most responses came from Nigeria (71.43%), with a sizeable share from the United Kingdom (28.57%), confirming that Tummy Clothing operates across different service expectations and logistical realities. Responses came from Lagos, Kano, Oyo, Rivers, Anambra, Abuja, London, Leeds, Enugu, Ogun, and FCT, giving the brand visibility across many environments at once.

A defining feature of the case was the multilingual nature of the feedback. Customers wrote in English, Nigerian Pidgin, Yoruba, and Hausa, often mixing them within a single message. This made the dataset richer but also more demanding to analyse. The BizIntel platform therefore had to read language in context rather than rely on literal wording, so that local customer voices were included in the collective intelligence process rather than excluded from it (Balahur & Turchi, 2014; Mabokela et al., 2023).

3.6 Chapter Summary

Taken together, the company's lean structure, multilingual customer base, and cross-border market make Tummy Clothing a strong site for collective intelligence research in an emerging market. Having established that context, the next chapter turns to methodology and system design.

Chapter 4: Methodology and System Design

4.1 Introduction

This chapter explains how the study was designed: from the deployment of the BizIntel feedback analytics platform, through the data collection phase, to the process of turning customer feedback into usable business insight. Because the research sits at the intersection of collective intelligence, customer experience, and applied analytics, a mixed-methods approach was the most appropriate way to capture both the numerical and the human sides of the problem (Reis et al., 2022; Venkatesh, 2023). The chapter also reflects a design-science orientation, since the research did not only observe a business from the outside but built and tested a working feedback analytics system inside a real SME (Hevner et al., 2004; Peffers et al., 2007).

4.2 Research Design

The study used a mixed-methods design because its questions required both measurement and interpretation. Quantitative methods captured metrics such as CSAT, NPS, and CES across the customer base, while qualitative methods explained the reasons behind those scores. This combination is valuable in business and management research because it keeps the analysis from resting on numbers alone or narrative alone (Reis et al., 2022; Venkatesh, 2023). The approach also fits the nature of collective intelligence, since customer feedback is not one type of data. It combines score metrics, free-text comments, emotional tone, and practical suggestions, and a single method would have been too narrow for such a dataset.

4.3 Design Science Logic

The study's methodological logic follows design science research. Design science suits situations where the researcher builds an artifact, in this case a digital solution, to address a

real-world problem and then evaluates how well that artifact works in context (Hevner et al., 2004; Peffers et al., 2007). Here the problem was the fragmented and underused state of customer feedback. The artifact was the BizIntel platform, which organised that feedback into structured intelligence. The platform was not counted as a success simply because it existed; it was evaluated on the quality of its output, the usefulness of its classifications, and how far it helped management identify operational issues. Three criteria guided that evaluation. The first was classification quality, meaning whether the sentiment labels and categories the platform produced matched what the comments expressed. The second was interpretive usefulness, meaning whether the aggregated outputs told management something it could act on rather than restating raw scores. The third was problem fit, meaning whether the artifact addressed the original difficulty of fragmented, underused feedback. Judging the platform against these three criteria, rather than against its mere existence, is what makes the design-science claim testable rather than assumed.

4.4 Data Sources

The study drew on three main data sources. The first was the customer feedback dataset collected through the BizIntel survey system, containing both numerical ratings and written feedback from Tummy Clothing customers. The second was the executive report generated from the platform, which summarised the key patterns in the data and produced the SWOT analysis and aggregated insight's view. The third was the business context itself: how the company operates, its customer base, and its feedback practices before and after deployment. The dataset included respondent location, gender, category of feedback, sentiment class, and LLM-generated summaries, which made it possible to examine both demographic distribution and thematic patterns. Taken together, the three sources triangulate: the raw dataset shows what customers said, the executive report shows how the platform organised those signals, and the

business context shows why particular patterns mattered for an SME operating across two markets. Each source covers a blind spot in the others, so that a claim supported across all three carries more weight than one resting on the platform output alone.

4.5 Sampling and Data Collection

The survey was distributed to active Tummy Clothing customers across three social channels: WhatsApp (35 recipients), Instagram (20 recipients), and Facebook (15 recipients), for a total distribution of 70 survey links. Customers were asked for consent before any link was shared. A total of 42 usable responses came in between 16 May and 30 June 2026, a response rate of 60 per cent (42 out of 70). That rate is strong for a social-media-distributed survey in an SME context and reflects the existing relationship between the brand and its customers (Chattermill, 2026).

The sample was not built to be statistically representative of the whole population of fashion consumers. It was built around actual users of the business during the research period, which suits a case study whose purpose is to understand a specific organisational setting in depth rather than to generalise broadly (Yin, 2014). The survey used a skip-logic structure, which lowered friction for respondents and encouraged fuller responses. Customers first scored their experience, then had space to explain it in their own words.

4.6 Survey Structure and Measures

4.6.1 Customer Satisfaction Score (CSAT)

Customer Satisfaction (CSAT) was measured on a five-point scale anchored at 1 (Very unsatisfied) and 5 (Very satisfied), with the survey question asking: 'How satisfied are you with the product and services?' CSAT scores of 4 or 5 were treated as favourable. Across the 42

responses, 27 respondents (64.29%) gave a CSAT score of 4 or 5, with a mean score of 3.52 out of 5. A mean of 3.52 sits above the scale midpoint but well short of the ceiling, and the spread matters more than the average: while roughly two-thirds of customers rated their experience favourably, the remaining third clustered at the lower end, signalling a segment whose dissatisfaction the qualitative feedback later explains in terms of sizing, quality, and response time.

4.6.2 Net Promoter Score (NPS)

Net Promoter Score (NPS) was measured on an 11-point scale from 0 (Never recommend) to 10 (Absolutely recommend), using the question: 'How likely are you to recommend this business's products and services to your friends and family?' NPS results are reported as the BizIntel platform computes them, treating 7 to 10 ratings as favourable. Under this convention, 28 respondents (66.67%) gave a score of 7 or above. For transparency, under the strict Reichheld (2003) convention, where only scores of 9 to 10 count as promoters and scores of 0 to 6 count as detractors, the NPS is approximately minus 12. The platform's favourable rate (66.67%) is used in the main report because it reflects the threshold defined in the BizIntel instrument design. The mean NPS score was 6.6 out of 10. The distance between the two figures itself is informative. A favourable rate of 66.67% under the 7-to-10 threshold alongside a strict score of approximately minus 12 indicates a customer base that is broadly willing to recommend but light on active promoters, the pattern typical of a brand that satisfies without yet delighting, which aligns with the qualitative finding that strong product appeal coexists with recurring service friction.

4.6.3 Customer Effort Score (CES)

Customer Effort Score (CES) was measured on a five-point scale anchored at 1 (Very Difficult) and 5 (Very Easy), using the question: 'How easy is it for you to get the products or services?' CES scores of 4 or 5 were treated as favourable. Across the 42 responses, 27 respondents (64.29%) rated the process as easy, with a mean CES score of 3.52 out of 5. The open-ended field let customers explain issues in their own language, and that explanatory detail is where the most operationally useful insight was found.

4.7 Data Cleaning and Preprocessing

Before analysis, the raw feedback had to be cleaned and prepared. The dataset held short comments, mixed-language text, informal expressions, and some variation in how people wrote their responses. Text preprocessing was carried out by the Gemini Pro large language model within the platform to normalise the data while preserving meaning. Research on sentiment analysis shows that preprocessing can improve classification but must be handled carefully so that important signals are not lost (Pradha et al., 2019; Dashtipour et al., 2016). The dataset was also arranged so that quantitative and qualitative fields could be processed separately where needed, preserving the integrity of each data type while still linking them in the final analysis.

4.8 LLM-Powered Interpretation

The platform used the Gemini Pro large language model to process free-text comments and classify them into meaningful categories, including sentiment labelling (Positive, Negative, or Mixed), thematic grouping, and short plain-language summaries for each entry. Gemini Pro's multilingual capability was used to interpret Nigerian Pidgin, Yoruba, Hausa, and code-switched expressions alongside English, reading meaning in context rather than counting words. Research confirms that LLM-based multilingual analysis works better when paired with

human judgment, especially in low-resource language settings (Balahur & Turchi, 2014; Koto et al., 2024).

4.9 Validation and Human Review

Gemini Pro handled the initial classification, but the research did not rely on automation alone. Human review formed a structured second stage, carried out in two complementary ways. The two stages served different purposes: the first tested whether the model's labels were accurate, and the second tested whether the negative signals it surfaced were authentic rather than artefacts of misclassification. Separating accuracy from authenticity in this way is what allowed the study to trust the platform's complaint volume as a basis for recommendations rather than treating it as unverified machine output.

First, the researcher read all 42 original customer comments alongside their Gemini Pro-generated classifications to check whether the sentiment labels and category assignments were accurate. Of the 42 entries, one was misclassified. A customer who had written in mixed Yoruba and English, expressing strong satisfaction and an intention to buy again, had been labelled Neutral by the platform. The original comment read: *'Omo, aso yi loud gan! The rich aunty boubou is exactly as seen on Instagram. Ese o, I will be buying more.'* The sentiment, from the exclamatory Yoruba praise to the explicit commitment to repurchase, was clearly positive. This entry was reclassified to Positive before final reporting, which changed the final sentiment distribution from 23 Positive and 1 Neutral to 24 Positive and removed the Neutral category from the dataset entirely. The correction shows why human review matters in multilingual settings: the model read the tonal register of informal Yoruba praise as affectively neutral, and the researcher's contextual knowledge was needed to catch it (Balahur & Turchi, 2014; Mabokela et al., 2023).

Second, all 10 entries classified as Negative were independently verified with a Tummy Clothing team member to confirm the authenticity of the complaints. The team members confirmed that every negative entry reflected a genuine issue customers had experienced and that the business had also observed internally, including sizing errors, fabric quality concerns, delayed deliveries, and slow customer service response times. No negative entries were disputed or removed. This triangulation between customer voice and internal operational knowledge strengthened the credibility of the findings and confirmed that the platform's complaint signals were grounded in real service failures rather than isolated misunderstandings.

4.10 Ethical Considerations

Ethics formed an important part of the methodology because the study involved customer feedback and contact-linked information. The most important concern was consent. Before any survey link was distributed, customers were approached individually on WhatsApp, Facebook, Instagram and were asked for their explicit agreement to take part. The process was direct and transparent: each customer was told their responses would be used for business improvement and research purposes, invited to respond YES if they agreed, and only then sent the survey link. Anonymous submission was also offered, so that customers who preferred not to be identified could still take part. Plate 1 shows an example of the WhatsApp consent request and the customer's YES response, followed by the survey link being shared.

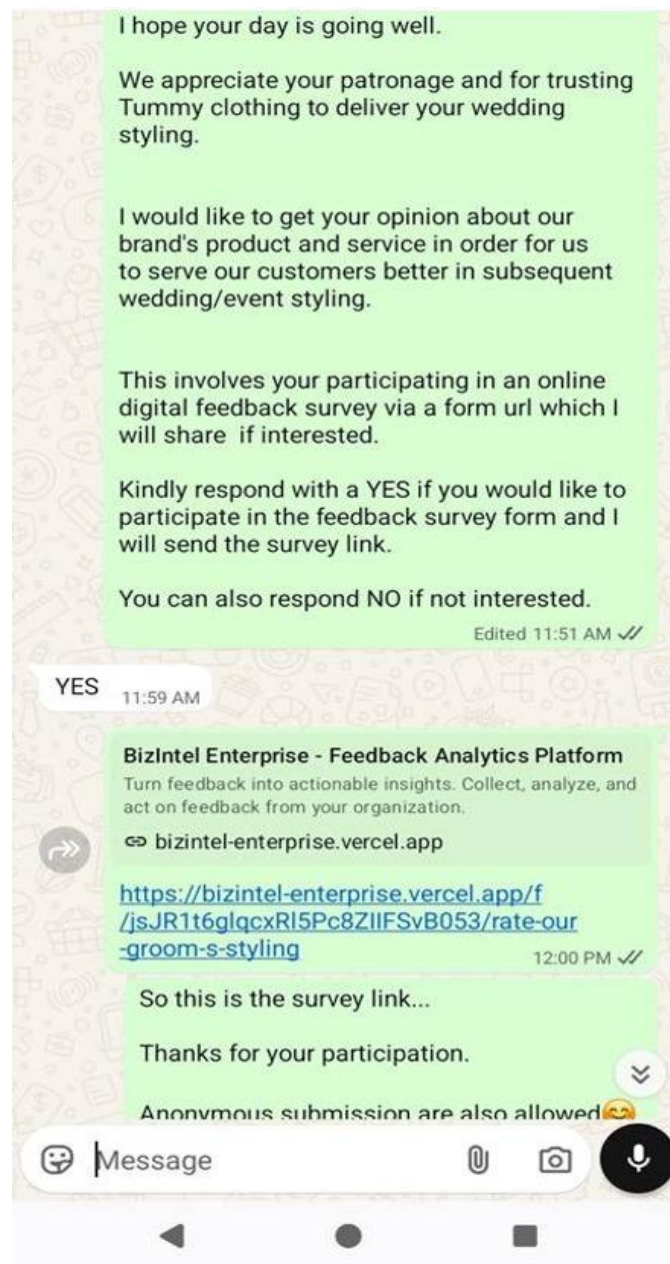


Plate 4: WhatsApp consent request and customer response confirming voluntary participation before the survey link was shared.

Personal data was handled securely so that respondents were not exposed unnecessarily. The study followed the principle of data minimisation: only information relevant to the research was collected and used, in line with the research ethics principles set out in the UM6P SCI Thesis Guidelines (2026). Customers who wrote in local dialects or informal language were not penalised for how they expressed themselves. Treating local speech as meaningful data rather

than noise helped maintain analytical integrity and supported a more accurate, inclusive reading of customer experience. Human oversight ran throughout: LLM-generated outputs were not treated as final truth without review, and the one misclassification identified during that review was corrected before reporting.

4.11 Chapter Summary

This chapter explained the study's methodology: the mixed-methods design, the design science logic, the data collection across three social channels at a 60 per cent response rate, the Gemini Pro-powered LLM analysis, the two-stage human validation process, and the ethical safeguards including the social media consent request protocol. The next chapter presents the findings and discusses what they mean for customer satisfaction, service quality, and managerial decision-making.

Chapter 5: Results, Findings, and Implications

5.1 Introduction

This chapter presents the main findings from the BizIntel deployment and discusses what they mean for Tummy Clothing as an SME. The results describe a business with real creative and product strengths but also clear operational friction around product quality, sizing, and customer service. Customer experience is best understood as the combined effect of product, service, and delivery rather than the outcome of a single transaction (Puccinelli et al., 2009; Verhoef et al., 2009). The findings follow that logic, combining the quantitative scorecards with the qualitative feedback themes, drawn from 42 customer responses whose sentiment classifications were corrected through researcher-led human review before reporting.

5.2 Demographic Analysis

The demographic profile of respondents gives important context for the findings. Knowing who the customers are, where they are, and how they divide by gender helps explain the variation in feedback themes and satisfaction levels across different parts of the business.

5.2.1 Country and City Distribution

Figure 1 presents the distribution of respondents by country. The data shows that 30 of the 42 respondents (71.43%) were based in Nigeria, while the remaining 12 (28.57%) were located in the United Kingdom. This two-market split confirms that Tummy Clothing is already functioning as a cross-border fashion brand. In customer experience terms, factors such as shipping reliability, delivery time, and return feasibility carry different weight depending on the customer's location (Verhoef et al., 2009; Puccinelli et al., 2009).

Location of Customers by Country (n=42)

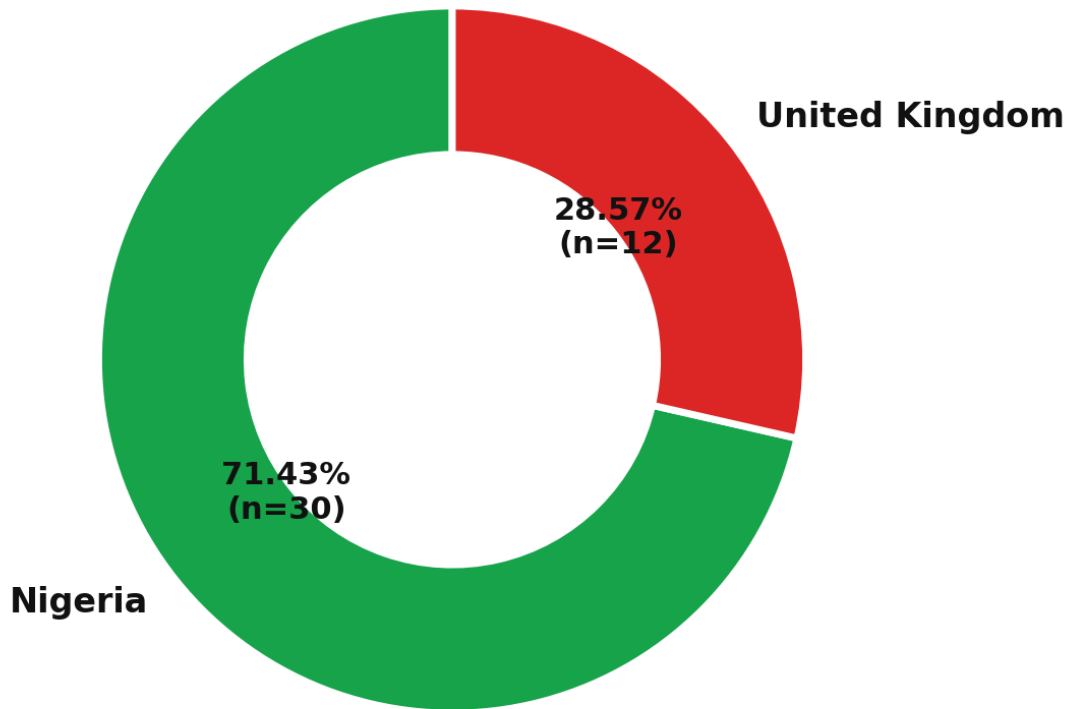


Figure 1: Location of customers by country: Nigeria (n=30, 71.43%) vs United Kingdom (n=12, 28.57%).

Figure 2 breaks down respondents by city. London recorded the highest single-city representation at 12 respondents (28.57%), making it the largest city-level group. Lagos followed as the second most represented city with 8 respondents (19.05%), consistent with the brand's origins and primary operational base. Abuja and Kano each accounted for 4 respondents (9.52%), while Enugu and Oyo each contributed 3 respondents (7.14%). Rivers (n=2, 4.76%), and Ogun, FCT, Leeds, and Anambra (n=1, 2.38% each) completed the distribution. This wide city-level spread confirms that the brand is managing service expectations across multiple environments simultaneously.

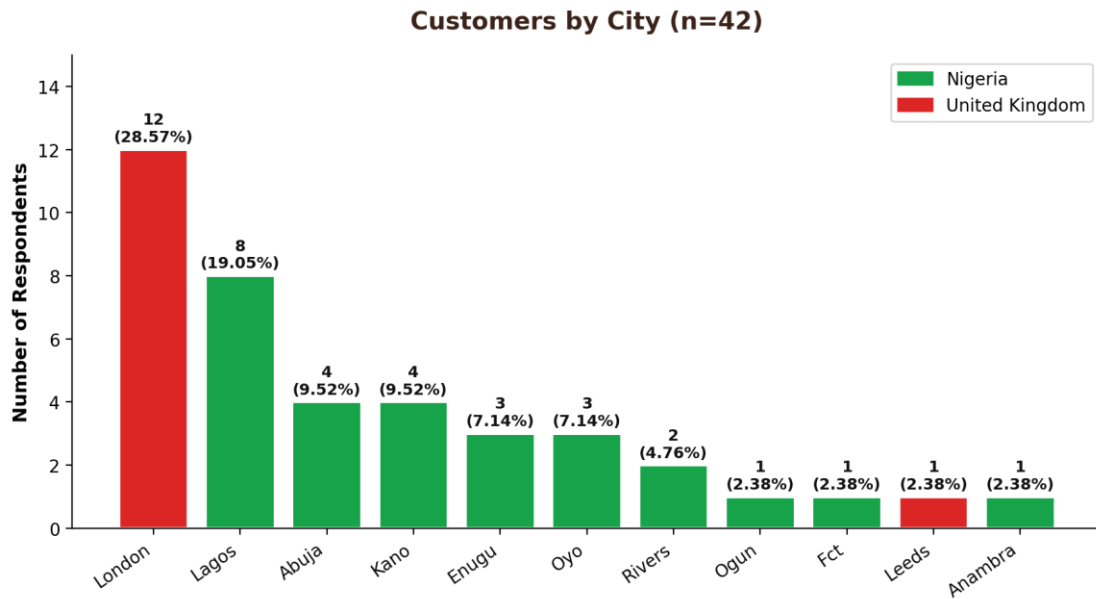


Figure 2: Distribution of customers by city: London (n=12, 28.57%), Lagos (n=8, 19.05%), Abuja and Kano (n=4, 9.52% each), Enugu and Oyo (n=3, 7.14% each), others below 5%.

5.2.2 Gender Distribution

Figure 3 presents the gender distribution and corrected sentiment classification across the full sample. Male respondents accounted for 25 of the 42 submissions (59.52%), female respondents for 16 (38.10%), and 1 respondent did not specify a gender (2.38%). The stronger male representation likely reflects the brand's particular strength in groom and occasion wear, which emerged as a recurring area of positive feedback (Kotler & Keller, 2016).

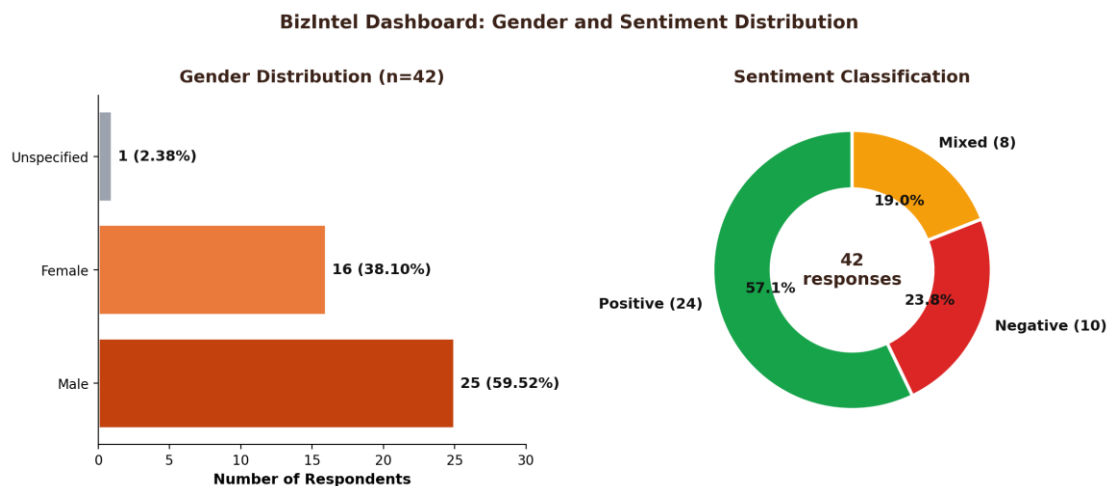


Figure 3: Gender distribution (Male: 25, Female: 16, Unspecified: 1) and corrected sentiment classification (Positive: 24, Negative: 10, Mixed: 8) across 42 responses.

5.3 Qualitative Findings

The qualitative findings help explain the numbers. The strongest positive themes concerned design quality, tailoring skills, fabric appearance, packaging, and successful delivery, especially for international customers. Customers repeatedly praised the fit, the presentation of the garments, and the sense that the brand understood their expectations. The clearest positive pattern was praise for bespoke tailoring and wedding wear. Plates 5 and 6 show how the platform interpreted both positive and negative customer voices, including those expressed in local Pidgin and informal Yoruba.

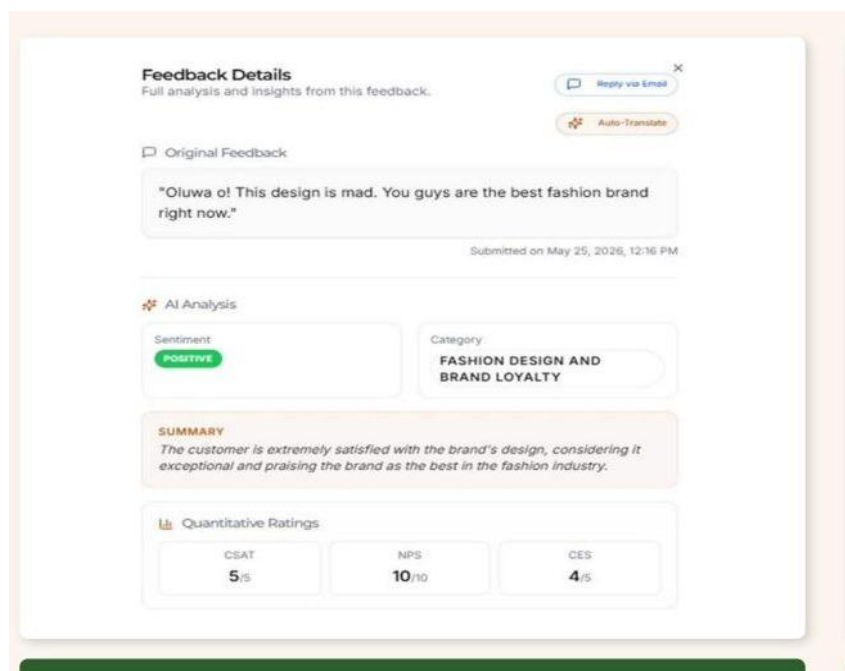


Plate 5: A customer praising Tummy Clothing's recent design in playful Yoruba-English language (CSAT 5/5, NPS 10/10).

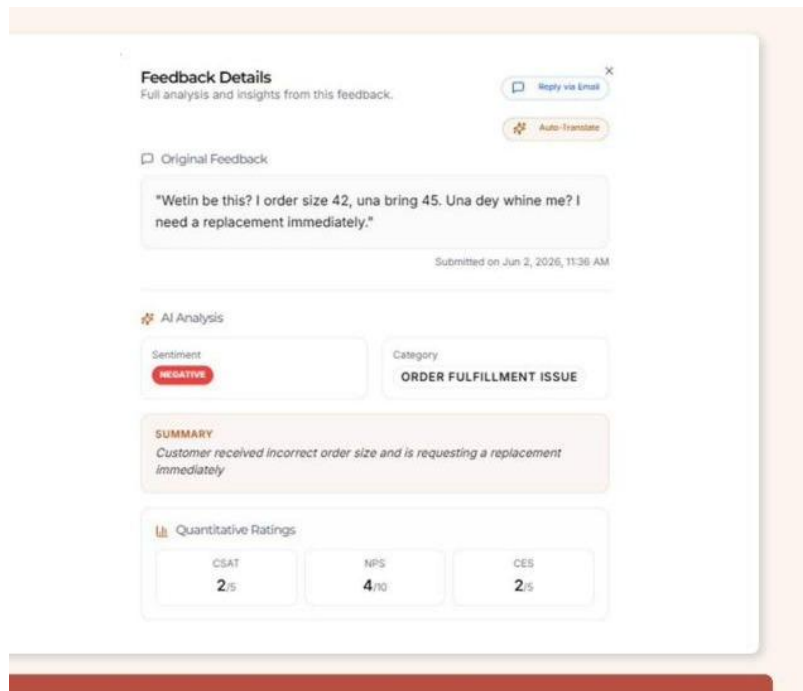


Plate 6: A customer expressing dissatisfaction with a recent order in Nigerian Pidgin (CSAT 2/5, NPS 4/10).

The negative themes were equally clear. The main complaints involved material fading, broken zippers, incorrect sizing, and slow customer service. All 10 negative entries were validated with a Tummy Clothing team member, who confirmed that each reflected a genuine and recurring operational issue the business had also observed internally. Figure 4 gives a visual breakdown of sentiment by feedback category. The Fashion and Design category drew the most responses overall (n=18) and was overwhelmingly positive (16 positive, 2 mixed, no negative entries). Product Quality and Sizing drew 9 responses with 4 negative entries. Delivery and Fulfilment drew 11 responses with 5 negative and 3 mixed entries, giving it the highest share of negative sentiment.

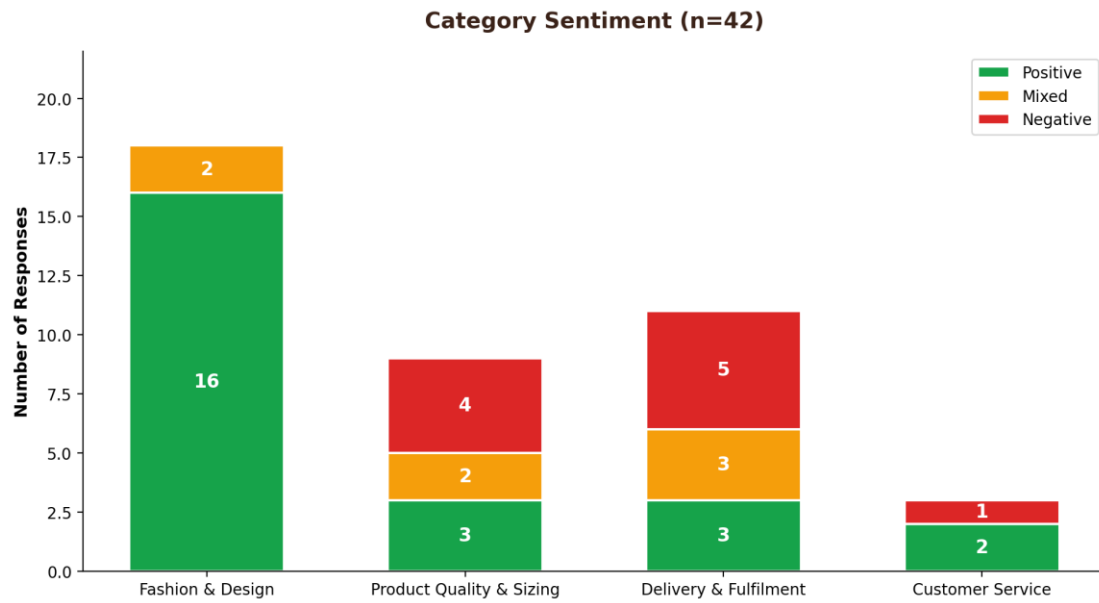


Figure 4: Category sentiment: positive, mixed, and negative counts across the four grouped feedback categories (n=42).

Figure 5 consolidates the sentiment data further. Following the researcher's correction of the one misclassified entry, the final distribution across the 42 responses is: 24 Positive (57.14%), 10 Negative (23.81%), and 8 Mixed (19.05%), with the Neutral category eliminated from the dataset after human review.

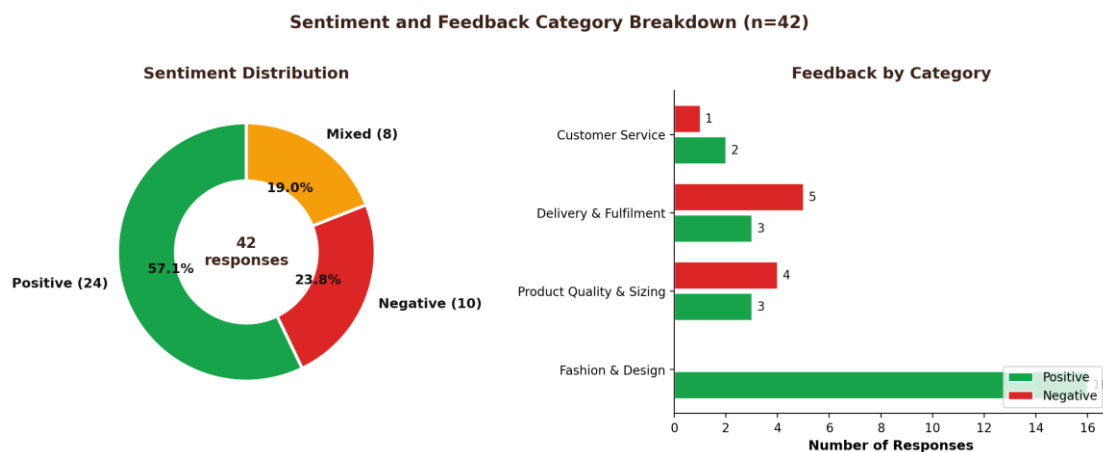


Figure 5: Corrected sentiment distribution (Positive: 57.14%, Negative: 23.81%, Mixed: 19.05%) and feedback by category across 42 responses.

5.4 Discussion of Key Themes

The first major theme is product strength. Customers responded warmly to the brand's tailoring skill, visual presentation, and garment aesthetics. The brand holds a valuable market position, held back not by the concept but by inconsistent execution in parts of the operation. Plate 7 presents the platform's aggregated intelligence view, which identifies inconsistent customer service as the most urgent pain point, with the strongest positive sentiment attributed to high-quality fabric and stitching. The platform also surfaced a hidden commercial opportunity: growing demand for specific popular items such as Ankara hoodies. Across the full dataset, the platform identified 10 complaints, 23 compliments, and 8 hidden opportunities. Read as a ratio, the 23 compliments against 10 complaints place positive sentiment ahead of negative by more than two to one, which reframes the brand's situation: the issue is not a weak product concept but uneven delivery of a concept customers already value. That distinction matters strategically, because problems of execution are more readily corrected through process discipline than problems of positioning, which would require rebuilding customer perception from the ground up.

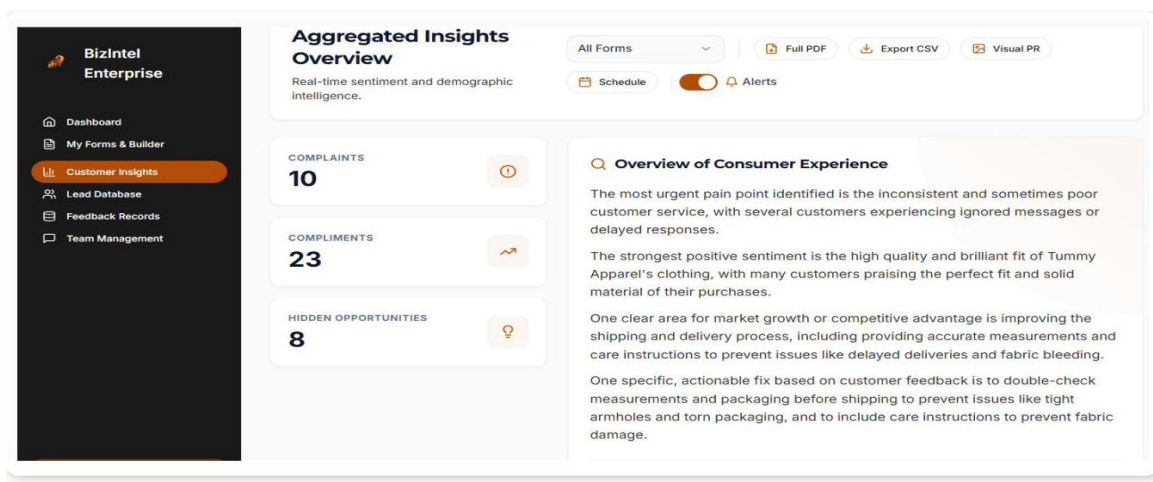


Plate 7: BizIntel aggregated insights overview: complaints (n=10), compliments (n=23), hidden opportunities (n=8), and LLM-generated actionable recommendations.

The second theme is quality control. Comments about fading fabrics and broken zippers point to problems with materials or supplier consistency. In retail, repeated product defects erode trust quickly because customers read them as signs of unreliability. The findings support the need for tighter vendor selection and stronger pre-dispatch checks.

The third theme is sizing accuracy, one of the clearest sources of dissatisfaction. In fashion, sizing problems are not minor; they affect returns, trust, and repeat purchasing. In international markets especially, incorrect sizing can be costly and discouraging for the customer.

The fourth theme is customer service response time. Several comments pointed to slow replies and delays in communication. This connects directly to CES, because a customer may like the product yet feel worn down by the process. In customer experience research, service responsiveness is among the factors that most shape loyalty and recommendation (Klaus & Maklan, 2013; Intercom, 2022).

5.5 Strategic Meaning of the Results

The results show that Tummy Clothing is a brand with strong product appeal and loyal customers, held back by operational inconsistency. Plate 8 presents the SWOT and Contention Analysis generated by the BizIntel platform.

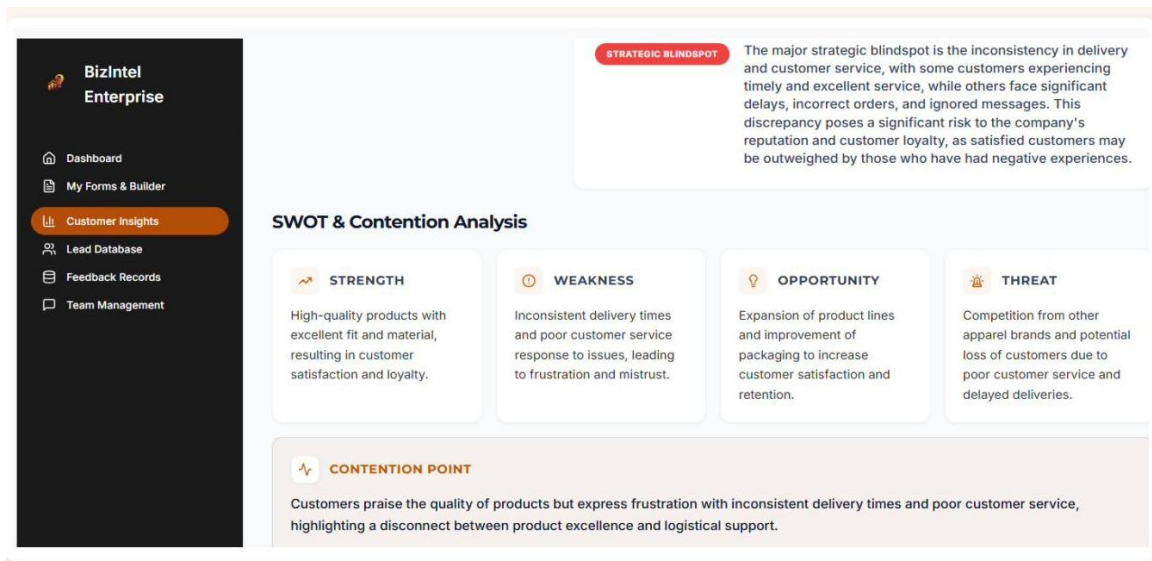


Plate 8: BizIntel SWOT and contention analysis: strength in tailoring quality, weakness in sizing and response time, opportunity in occasion wear demand, threat from reputational risk

The Strength quadrant confirms that high-quality fabrics and excellent tailoring are the brand's competitive core. The Weakness quadrant identifies inconsistent sizing and slow customer service response as the main operational liabilities. The Opportunity quadrant highlights growing demand for traditional and occasion wear. The Threat quadrant flags competition from other fashion brands and the reputational risk created by recurring quality control failures. The Contention Point ties these together: while many customers praise quality, others voice frustration with sizing and slow service, exposing a disconnect between the brand's strengths and its operational weaknesses.

The findings also support the logic of collective intelligence. Read individually, some issues look minor. Aggregated across 42 responses and reviewed with human judgment, a clearer story emerges. Customers value the brand's creative output but want better reliability in quality control, sizing, and communication. That pattern gives the business a clear, evidence-based direction for action (Woolley et al., 2015; CoachHub, 2026).

5.6 Discussion in Relation to Literature

The findings sit comfortably with the literature on customer experience in retail. Retail scholars have argued that the buying journey matters as much as the product itself, and this study confirms that view in a real SME (Verhoef et al., 2009; Puccinelli et al., 2009). Customers did not judge Tummy Clothing only by the garment they received; they judged it by how easy it was to communicate, how long they waited, and how closely the product matched their expectations.

The findings also align with research on feedback analytics and collective intelligence. The platform's category sentiment charts, SWOT analysis, and aggregated insights view organised what might have stayed buried in 42 individual comments into visible, recurring themes (Chattermill, 2026; Qualtrics, 2025; Woolley et al., 2015). The multilingual component and the one misclassification caught during human review reinforce the literature on under-resourced language analysis. The platform processed 41 of 42 entries across multiple languages without human intervention, and the one Yoruba exception shows why human oversight belongs in any such system (Balahur & Turchi, 2014; Mabokela et al., 2023).

5.7 Chapter Summary

The quantitative scores placed the business in broadly positive territory, with 64.29% of customers giving favourable CSAT and CES scores and 66.67% giving a favourable NPS under the platform's 7-to-10 convention. The corrected final sentiment distribution is 24 Positive (57.14%), 10 Negative (23.81%), and 8 Mixed (19.05%). The qualitative data shows where the gap remains: design and tailoring generate real enthusiasm, while quality control, sizing, and response speed still disappoint. That gap between product strength and operational

inconsistency is the central finding of the chapter and sets up the recommendations in Chapter 6.

Chapter 6: Recommendations and Conclusion

6.1 Summary of the Research

This study sets out to examine how collective intelligence and feedback analytics can improve decision-making in a fashion SME. The central argument was that customer feedback deserves to be treated as an active business resource, one that can reveal strengths, weaknesses, opportunities, and service bottlenecks, rather than just a random record of opinions. The case of Tummy Clothing showed that the idea is not merely theoretical: it can be implemented in real business and used to guide action (Woolley et al., 2015; CoachHub, 2026).

The research combined relevant customer experience evaluation scoring metrics with multilingual written feedback to understand both what customers felt and why. That mixed approach mattered because customer experience in retail is shaped by the full journey, including product quality, delivery, communication, and post-purchase support (Verhoef et al., 2009; Puccinelli et al., 2009). The BizIntel platform made it possible to organise this data into patterns management could actually use. The two-stage human review confirmed the value of combining LLM output with researcher judgment: one misclassification was identified and corrected, and all 10 negative entries were independently verified with a team member, strengthening the credibility of the findings.

The findings also confirmed the value of multilingual feedback analysis. Customers expressed themselves in English, Nigerian Pidgin, Yoruba, and Hausa, and Gemini Pro interpreted those voices meaningfully in 41 of 42 cases. The study therefore contributes to both practice and literature by showing how local language can serve as part of strategic intelligence rather than being treated as a limitation (Balahur & Turchi, 2014; Mabokela et al., 2023).

6.2 Key Findings

The strongest finding is that Tummy Clothing has a clear product strength. Customers repeatedly praised the tailoring, design, fabric appearance, and packaging. The brand already holds a strong creative identity and a solid base of customer trust, a major advantage in fashion retail (Puccinelli et al., 2009).

The second major finding is that the organisation has operational weaknesses in quality control, sizing accuracy, and customer response speed. All 10 negative entries were confirmed as genuine recurring issues through internal team verification, giving the findings credibility beyond the platform output alone (Klaus & Maklan, 2013; Intercom, 2022).

The third major finding is that the platform made hidden opportunities visible. Some customers asked for restocking of specific items, while others expressed strong interest in repeat purchases. This suggests the brand holds demand signals it can use for production planning and inventory decisions, turning collective intelligence into commercial growth (Woolley et al., 2015; Chattermill, 2026).

6.3 Practical Implications

The study carries several practical implications for Tummy Clothing. The business should treat customer feedback as a routine management tool rather than a one-time report, reviewing the BizIntel dashboard regularly so that customer trends are not missed. It should formalise quality control, since the complaints about fading fabric and broken zippers point to material choice and inspection procedures that need attention. It should improve sizing support through a digital sizing guide, clearer measurement instructions, and stronger pre-order consultation. And it should improve communication speed, because responsiveness is now part of the experience itself (Klaus & Maklan, 2013; Intercom, 2022). Framed together, these implications point

toward a single discipline: treating each recommendation as a change whose effect can be measured on the same CSAT, NPS, and CES instruments used here, so that the platform becomes not only a diagnostic tool but a way of tracking whether operational fixes actually change customer sentiment and improve their overall experience over time.

6.4 Recommendations for Tummy Clothing

The first recommendation is to institutionalise feedback review. Management should make customer feedback a standing item in business review meetings, moving from reactive problem-solving to anticipatory governance. The BizIntel dashboard should be reviewed on a regular cycle so that customer trends are not missed and emerging issues are addressed before they affect loyalty (Woolley et al., 2015; CoachHub, 2026).

The second recommendation is to strengthen production quality controls. Tummy Clothing should review its material suppliers, stitching checks, and final inspection process. Products intended for premium or occasion wear must meet high durability expectations and have better quality control that will protect the brand and reduce returns.

The third recommendation is to improve sizing standardisation. The company should develop a clear measurement guide for customers, especially international buyers who cannot easily correct errors in person. It should also standardise its consultation process so that measurements are recorded and checked before production begins.

The fourth recommendation is to expand customer service capacity. The brand should ensure customer messages are answered quickly and consistently whether by assigning a dedicated staff member to customer communication or by setting clear response-time targets (Intercom, 2022; Chattermill, 2026).

The fifth recommendation is to use feedback for demand planning. Requests for specific products should be read as meaningful market signals. Rather than waiting for sales alone to determine what to produce, the company can use customer requests as an early indicator of demand, reducing inventory uncertainty and focusing production on products with clear customer interest.

6.5 Limitations of the Study

This study has several limitations, though each is bound by the design choices behind it. The first is that it rests on a single business case. Tummy Clothing is a valuable case, but the results cannot automatically be generalised to all fashion businesses or all SMEs; consistent with the case-study logic set out earlier, the aim was analytical depth within one organisational setting rather than statistical generalisation across a population, so the findings are best read as transferable patterns to test elsewhere rather than universal claims. The second is the time frame: the research covers the deployment period from May to June 2026, so the findings reflect a particular stage in the company's operations and may not capture seasonal or longer-term patterns, which repeated deployment across several quarters could address. The third is the reliance on LLM interpretation. Gemini Pro handled mixed-language feedback well, but one misclassification was identified in the dataset, and no automated model is perfect with sarcasm, idiom, and local nuance; the two-stage human review was built into the design precisely to contain this risk, and it caught the single error before reporting. The fourth is that some feedback may reflect self-selection bias, since survey respondents are often those with strong opinions in either direction. Even so, the combination of quantitative scores, qualitative comments, and two-stage validation makes the findings robust for understanding broad trends in customer experience.

6.6 Future Research

Future research could extend this work in several directions. One would apply the BizIntel approach to other SMEs in retail, hospitality, or services to test whether the same collective intelligence model holds in different business environments. Another would examine the long-term effect of the recommendations made here, moving from diagnosis to impact evaluation by tracking whether CSAT, NPS, and CES scores improve after quality control changes, sizing standardisation, and customer service improvements are put in place. A third would explore more advanced multilingual models for under-resourced language environments, since the one misclassification in this study involved Yoruba, and future work could improve classification accuracy by testing stronger language models or hybrid human-LLM systems (Balahur & Turchi, 2014; Mabokela et al., 2023).

6.7 Concluding Remarks

This research confirms that customer feedback is a strategic asset, not a post-purchase formality. When businesses have the tools to collect, interpret, and act on that feedback, they can improve product quality, service quality, and customer loyalty. The BizIntel platform showed that collective intelligence can be made practical for an SME by turning many voices into clear operational guidance. Tummy Clothing already has strong creative talent and customer goodwill; what it needs now is greater consistency in quality, communication, and sizing. With the right system and the discipline to use it regularly, the brand can strengthen its position and grow more sustainably. More broadly, the study shows that SMEs need not remain dependent on guesswork. Given a system that captures what customers are saying, they can replace it with evidence.

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